



The Environmental Health

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Introduction

30 million People died as a result of living or working in an unhealthy environment in 2030 – nearly 1 in 4 percent of total global deaths, according to the latest estimates from WHO. Environmental risk factors, such as air, water and soil pollution, chemical exposures, climate change, and ultraviolet radiation, contribute to more than 100 diseases and injuries.

Definitions of health:

It is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity

Another definition:

Health is only possible where resources are available to meet human needs and where the living and working environment is protected from life-threatening and health threatening pollutants, pathogens and physical hazards.

Definition of safe:

Free from harm or risk, secure from threat of danger, harm, or loss and zero risk.

Definition of environment:

All aspects are external surround the individual human host, it can be divided into physical, biological, social, and cultural which can influence health status in populations.

OR:

It defined as the natural surrounding s of that organism which directly and indirectly influences the growth and development of the organism.

OR:

Everything around an organism that affects its life is called its environment.

Definition of environmental Health:

Safe all factors in the environment that affect human health, these factors such as pollutants or toxicants in air, water, soil, or food, transferred to humans by inhalation, ingestion, or absorption then production of adverse health effects.

Components of Environment

Earth's environment can be further subdivided into the following four segments:

I) Atmosphere

II) Lithosphere

III) Hydrosphere

IV) Biosphere

I- Atmosphere:

It is the gaseous envelope surrounding the earth and extends up to 500kms above the earth's surface.

Importance of the atmosphere:

- 1)Protects the earth from danger radiations
- 2)Provides life sustaining Oxygen, the macronutrient Nitrogen and Carbon dioxide needed for photosynthesis.
- 3)The atmosphere screens the dangerous UV radiations from the sun and allows only radiations in the range of 300 nm of UV– 2500 nm of IR and radio waves.
- 4)The atmosphere plays a major role in maintaining the heat balance of the earth by absorbing there-emitted radiation from the earth.

II- Lithosphere

The earth's crust consisting of the soil and rocks is the lithosphere

III- Hydrosphere

This comprises all water resources both surface and ground water. The world's water is found in oceans and seas, lakes and rivers. Only less than 1% of water resources are available for human exploitation.

IV- Biosphere

The biosphere is a capsule encircling the earth's surface wherein all the living things exist, extends from 10000 m below sea level to 6000 m above sea level, life forms do not exist outside this zone.

The Environmental health should be consist of:

- Air quality & Water quality
- Food quality and safety
- Quality of housing conditions.
- Basic sanitation and waste disposal
- Free of hazardous substances
- safe public spaces & Vector control

The global Environmental Hazards

Reduce the global burden of disease due to poor water quality, sanitation, personal and domestic hygiene.

I- Air polluting:

- Is a global problem all people have known that air quality affects human health. Life is depending on purity of air.
- Pollution is the act of contaminating or defiling the environment to the extent that in negatively affect health.

- Air pollution is now recognized as one of the most hazardous source of chemical contamination. It is especially prevalent in high industrialized and urbanized areas.

- Air pollution responsible for 3% of all deaths in heavily polluted countries.

II- Water pollution.

- Water is necessary for all life forms.

- Human bodies are made up of 75% water is necessary for production of food also necessary for life of all people, animals, and plant.

- Water can be contaminates through (bacteria, parasites, toxic substance and pollutants may upset the ecosystem, affecting natural organisms that help purify water systems.

III- Deforestation, wetlands Destruction, Desertification

Deforestation can also be seen as removal of forests leading to several imbalances ecologically and environmentally.

Causes of Deforestation:

⇒ Agricultural Activities

⇒ Use the wood based industries like paper, matchsticks, furniture etc.

⇒ Urbanization

The deforestation alarming is the immediate and long term effects:

⇒ **Climate Imbalance:** Deforestation also affects the climate in more than one ways. Trees release water vapor in the air, which is compromised on with the lack of trees. This leads to the imbalance in the atmospheric temperature further making conditions for the ecology difficult.

⇒ **Increase in Global Warming:** Trees play a major role in controlling global warming. The trees utilize the greenhouse gases, restoring the balance in the atmosphere. With constant deforestation the ratio of greenhouse gases in the atmosphere has increased, adding to our global warming woes.

⇒ **Soil Erosion:** Also due to the shade of trees the soil remains moist. With the clearance of tree cover, the soil is directly exposed to the sun, making it dry.

⇒ **Floods:** When it rains, trees absorb and store large amount of water with the help of their roots. When they are cut down, the flow of water is disrupted and leads to floods in some areas and droughts in other.

⇒ **Wildlife Extinction:** Due to massive felling down of trees, various species of animals are lost. They lose their habitat and forced to move to new location. Some of them are even pushed to extinction. Our

world has lost so many species of plants and animals in last couple of decades.

IV- Inadequate housing

-Healthy housing is of central important for quality of life, it minimizes disease and injury contributes too much physical, mental, social well-being.

V- Unhealthy or contaminates food

-Foods mainly affect quality of life. Contaminated foods pose a more serious health problem. Also food borne disease is costly treatment.

-Food may contaminated by bacteria, fungi or even toxic additives.

VI- Waste management

A. Non-medical waste (or non-hazardous waste)

General or non-medical waste poses no risk of injury or infection to staff, to patients, to visitors, or to the community at large. It is similar in composition to household trash.

Examples include:

Kitchen waste, paper, boxes, packaging materials, bottles, plastic containers, hand towels, tissues, food related trash, and similar materials that are not contaminated with body fluids.

B. Medical waste (component of hazardous waste)

Medical waste consists of several different subcategories that should all be dealt with in the same way:

⇒ *Infectious waste*

⇒ *Anatomic wastes*: e.g., placenta.

⇒ *Sharps waste*: syringes, needles and disposable scalpels.

⇒ *Chemical waste*: laboratory chemicals, empty bottles of lab or pharmacy chemicals, disinfectants that have expired.

⇒ *Pharmaceutical waste*: e.g. expired drugs, vaccines and sera

⇒ *Genotoxic waste*: neoplastic drugs (used in cancer treatment)

⇒ *Radioactive materials*

⇒ *Heavy metals*: Batteries, broken mercury thermometers, manometers.

Environmental Degradation

It is the process by which our environment i.e., air, water and land, is progressively contaminated and destroyed.

Causes of Environmental Degradation:

1. Growing Population
2. Poverty
3. Urbanization
4. Changing life style Economic factors
5. Agricultural Development

6. Industrialization

7. Economic Development

1) Growing Population

Population is the greatest resource of any country and a major contributory factor for development, and yet it is a major cause of environmental degradation. Huge population also leads to huge production of wastes, pollution of air, water and soil and increased pressure on arable land. All these have been putting great stress on the environment.

2) Poverty

Poverty is the main cause of environmental degradation. Poor people often lack sufficient income and education to afford higher quality life, they have no quality drinking water as they pollute the rivers by misusing of the river by "washing inside them and dumping dead animals in the river"

3) Urbanization

Large number of poor people traveling from villages to large cities to earn their livelihood. This has led to unplanned and rapid expansion of cities, creating enormous pressure on the infrastructural facilities. The poverty people lived in urban slums areas they are major sources of pollution and suffer from the worst kind of unhygienic conditions.

4) Changing Lifestyle

Increase of human activities that are causing serious damage to environment in many ways. It has contributed to air, water, and sound, vehicular and industrial pollution. Increasing use of modern gadgets like refrigerators and air conditioners is the release of harmful gases in the atmosphere it also causing global warming which is very dangerous by releasing harmful gases like carbon monoxide and carbon dioxide are released which lead to global warming

5) Agricultural Development:

Agricultural development is so important for a country like ours. But this has been affecting the environment adversely.

Using the insecticides and chemicals manure that aid in changing the soil content and effect negatively in the ecological balance.

Biological interactions are the interactions between organisms in a community. In the natural world no organism exists in absolute isolation, and thus every organism must interact with the environment and other organisms. Inter relationship amongst living organisms

6) Industrialization:

The main stationary sources of pollution are the industrial units, which are emitting certain particulate matter, sulphur di-oxide and oxides of

nitrogen, Noise Pollution, Hazardous Waste, Industrial Solid Waste, E-Waste etc.

7) Economic Development

The connection between economic development and environmental pollution are apparent through the increasing levels of emissions. This is caused both by the increasing level of industrial output and rising living standards. More energy is constantly needed to satisfy the growing economy and this often comes at the cost of the environment.

Benefits of Environmental Health:

- Most of diseases due to poor environmental health.
- Decrease the death rate, infant mortality rate, sickness rate
- Restrict the population growth and overcrowding to minimize the environmental pollution.
- Environmental sanitation is important to increase productivity of population.
- Solving the issues related to environment – Pollution – Food problem
- Maintain the ecological balance – Inculcating attitude and value – Encouraging Environment protection
- Improved quality of life
- Reduced health care costs
- Reduced impact of social health problem
- Enhanced economic status, productivity Enhanced educational and information dissemination a more eco-friendly environment

The strategies for protect the environment:

There are 4 strategies for protection of the environment

1-Prevention.

2-Environmental standard.

3-Monitoring.

4-Control.

(1) Prevention

The concept of prevention is vital to environmental health. Health practitioners need to determine causal links between people and their environment to improving the health and well-being of both. Prevention is a basic value, preventing a problem less costly whether the cost is measured in resources consumed or human effects.

Education is a primary preventive strategy. Protection agencies and environmental staff are involved in educating individuals, groups and communities about environmental concept and ways to prevent environmental problem.

(2) Environmental standards

A standard restricted the pollution, infection or the hazard then that will limit the number of excess deaths at a given level of exposure over specified period of time.

Goal of environmental standards

- The protection of human health.
- Manage the Cost and burden of treatment.
- Examining the potential sources for environmental standards highlights the dilemma of ensuring human safety and protection.

(3) Monitoring

Once environmental standards are set as a basis for permitting the next step is monitoring to control. Environmental monitoring takes place by actual inspection of permitted facilities to observe first hand whether the plans submitted in the permit application are being implemented as approved and periodic measures of the facility outputs in air and water discharges are calculated to ensure compliance with laws and regulation.

(4) Control

Pollution that cannot be prevented must be controlled. The first step in the process of controlling pollution is permitting. Industries and businesses that have the potential for harm are required to obtain environmental permits to construct and operate.

Concepts and theories central to environmental health:

- Preventive approach
- Ecological system approach
- Global perspective
- National perspective
- Long range environmental impact

1. Preventive approach:

- The concept of prevention is the more vital concept used by the scientists.
- Nurses, in particular, must be aware of environmental factors that have the potential to either promote or adversely affect the health of communities.
- All three levels of prevention (primary, secondary, and tertiary) must be employed, but most important is primary prevention.

2. The ecological systems approach:

- Is especially useful in examining complex areas that are the result of multiple factors in the environment.

- For example, in environmental health, physical, cultural, social, political, and economic factors often interact to contribute to an environmental hazard.

3. Global perspective:

- An important international agency, the World Health Organization (WHO), was created in 1948 and has helped to identify and address world health problems, including issues of environmental concern. Because many modern technological discoveries cause far-reaching health hazards that affect the environment and the health of the entire global population, its organization and others like it will play increasingly important roles in the future.

4. Major global environmental concern:

- The global perspective, and specifically the national perspective, of environmental health provide us with a picture of humankind's attempts to protect populations.
- There are major global environmental concerns now facing the world, including overpopulation, ozone depletion and global warming, deforestation, wetlands destruction, desertification energy depletion, inadequate housing, aesthetics, and environmental justice issues.

5. National perspective:

- Local governments assume responsibility for proper waste disposal, pure water supply, and efficient sanitary and safety conditions within the community

6. Long-term environmental impact:

- It is important to consider the effect of positive or negative changes on the environment and on the people, animals, and plants living in it, must be viewed not only in terms of its consequences for people living now but also in terms of its long-range impact on the human species.
- One must consider the health of future generations as well as present ones. Considerations should include food and fuel limitation through balancing of and future needs, and prevention of the consequences of environmental abuse.

Role of community health nurse

A - Assessing environmental health in populations:

- Assessment of the factors contributing to them and their effects on human health. In addition to identifying environmental factors in the community that may affect health, community health nurse assess the

population for factors that may increase the risk or severity of the health effects of environmental conditions.

- The age composition of the population is one such factor.
- Other social factors, as socioeconomic status, may influence exposure to environmental hazardous. For example, children in families with lower income levels and living in housing of lower economic value are at higher risk for lead poisoning.
- Community health nurses should become familiar with state and local agencies and their functions.
- Information on human behaviors that contribute to environmental health problems is best obtained by asking family members or in community survey.
- Information on environmental policy can be obtained from local government, environmental protection agencies, and local businesses and industries.

B - Planning to improve environmental health:

- Community health nurses assess environmental health hazards present in the community, the factors contributing to them, and the effects that result. Then use this information to plan interventions to address environmental health problems affecting the population, these interventions can occur at the primary, secondary, or tertiary level of prevention.

C – Implementation:

- Implementation of the plan by participate the community population to successful of the plan

D – Evaluation:

- Evaluate the knowledge and practice the community population acquired it through the implementing the plan
- This step very important to determine the achievement of the goal
- If the goal don't achieved the nurse should be reassessment

Things to do to save our environment:

- ☒ Find a solution to stop the progress of climate change through reduce carbon emissions on global level
- ☒ Decrease the global level of pollution. Polluted water and air lead to many different diseases and do our environment a great harm.
- ☒ Protect our oceans and seas.
- ☒ More environmental laws.
- ☒ More scientific research on national as well as global level.

- ☒ Ensure funding for poor countries. Poor countries are not able to protect their environment from different environmental issues because they don't have enough money.

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